

BEEP UTILITY IN C AND VISUAL C++



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Many people still use PCs and laptops without sound cards. Such a PC can't sing, but that doesn't mean it lacks the capabilities—it has to be instructed. PCs have an internal speaker, which is capable of generating beeps of different frequencies through software program instructions.

Two techniques are described here to make the internal speaker work for a DOS-/Windows-based PC without a sound card. These programs can also be useful if you can't access the external speaker for whatever reason. The internal speaker becomes an important component for testing various applications. For example, if you are developing an embedded system application where you want to include an error beep whenever error occurs, this program will come handy.

The internal speaker of your PC is the most primitive sound system used in IBM-compatible PCs. It was the only sound device available to PC games before advanced sound cards were introduced late in the 1980s. However, sound cards originally did not follow any common standard and were largely incompatible with each other. The internal speaker was the only sound system that could be regarded as universally present and used for various audio related applications.

How it works?

Music can be generated on your PC by sending a rapid series of musical notes to the speaker. The pitches for common musical notes up to six octaves are listed in the table on the next page. The notes are designated by seven alphabets, viz, A, B, C, D, E, F and G.

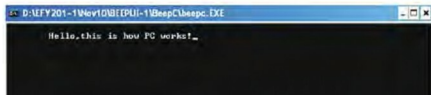


Fig. 1: Screenshot of the beep.c program output

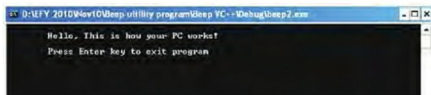


Fig. 2: Screenshot of the beep2.cpp program output

There are also 'H' or 'b' notes. For example, 'A#,' pronounced as 'A sharp,' is equivalent to 'Bb,' pronounced as 'B flat' note.

The 'A' note in octave 4 warbles at a frequency of 440 Hz. To play that note, send a series of signal pulses to the PC's speaker at a rate of 440 times per second. In other words, you control the speaker by providing a frequency number, which determines the pitch of the beep. A higher frequency number will cause the pulses to be sent quicker, resulting in a higher pitch.

The PC's speaker is best characterized by its inability to play more than one tone at a time. In spite of its limitations, the PC's speaker was often used in very creative ways to make the impression of polyphonic music or sound effects. It requires some effort, creativity and patience to produce a good sound on the available medium.

Several programs could play pulse-code-modulated (PCM) sound through the internal speaker using special techniques. However, because the method used to reproduce PCM was very sensitive to timing issues, these effects

either caused noticeable sluggishness on slower PCs or sometimes failed completely on faster PCs.

All of the latest operating systems include a generic sound application program interface (API), so most applications no longer need to know the specifications of each sound card. Correspondingly, the use of high-quality sound hardware has become commonplace. As a result, the internal speakers today are mainly used for warning, alert and start-up error beeps only, even though they can still be utilised for many audio applications. However, an onboard miniature piezo speaker's acoustic properties are significantly different from the older paper cone speakers', so most of the more complex sound generation techniques do not work.

Software program

Two programs are described here to make the internal speaker work with the help of various C and C++ built-in functions.

The beep.c is a simple 'C' program compiled using Turbo C IDE. When you run the program, you will see "Hello, this is how your PC works!"